Ch 4: The Market Forces of Supply and Demand

Mankiw, Principles of Microeconomics, Chapter 4

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EC 111 Principles of Microeconomics, Bentley University, Fall 2026

This chapter

- What determines how much buyers want to buy: demand
- What determines how much sellers want to sell: supply
- How a price emerges from their interaction: equilibrium
- How the price responds when something changes: comparative statics
- Why the price is the signal that allocates resources

Two lectures. Lecture 1: demand and supply. Lecture 2: equilibrium and changes in equilibrium.

Demand and supply

Markets and competition

Market

A group of buyers and sellers of a particular good or service. Buyers determine demand; sellers determine supply.

Perfectly competitive market

- Many buyers, many sellers, each too small to move the price
- Sellers’ goods are identical
- Everyone is a price taker: takes the market price as given, chooses only how much to buy or sell

Few markets fit exactly (wheat comes close). We start here because it is the simplest case and the logic carries over.

Demand

Quantity demanded

The amount buyers are willing and able to purchase at a given price.

Law of demand

Other things equal, quantity demanded falls when the price rises and rises when the price falls.

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The amount buyers are willing and able to purchase at a given price.

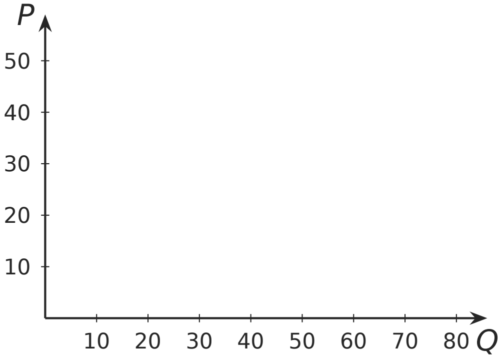
Law of demand

Other things equal, quantity demanded falls when the price rises and rises when the price falls.

“Other things equal”: income, prices of other goods, tastes, expectations, number of buyers are all held fixed. We release them one at a time later.

Demand schedule and demand curve

Price ($)	Q_d
0	50
10	40
20	30
30	20
40	10
50	0



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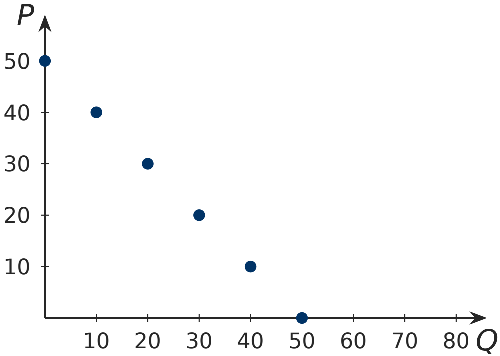

Demand schedule and demand curve

Price ($)	Q_d
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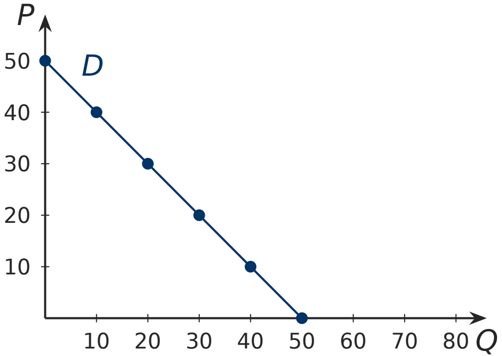
20	30
30	20
40	10
50	0

$Q_d = 50 - P$

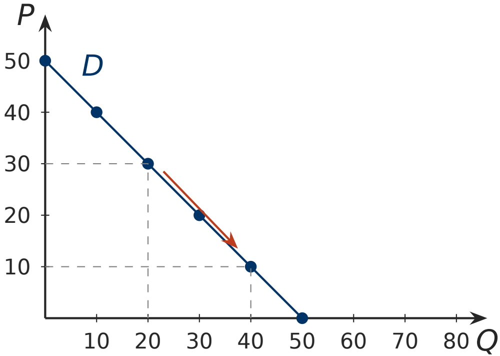


Demand schedule and demand curve

Price ($)	Q_d
0	50
10	40
20	30
30	20
40	10
50	0

$Q_d = 50 - P$

A price cut from $30 to $10 raises Q_d from 20 to 40: a movement along the curve.



Individual demand and market demand

Market demand is the sum of individual demands: at each price, add up what every buyer wants.

Price ($)	Buyer A	Buyer B	Market Q_d
0	30	20	
10	24	16	
20	18	12	
30	12	8	
40	6	4	
50	0	0	

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Price ($)	Buyer A	Buyer B	Market Q_d
0	30	20	50
10	24	16	40
20	18	12	30
30	12	8	20
40	6	4	10
50	0	0	0

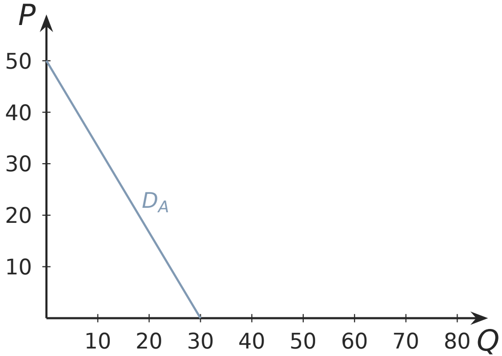
Individual demand and market demand

Market demand is the sum of individual demands: at each price, add up what every buyer wants.

Price ($)	Buyer A	Buyer B	Market Q_d
0	30	20	50
10	24	16	40
20	18	12	30
30	12	8	20
40	6	4	10
50	0	0	0

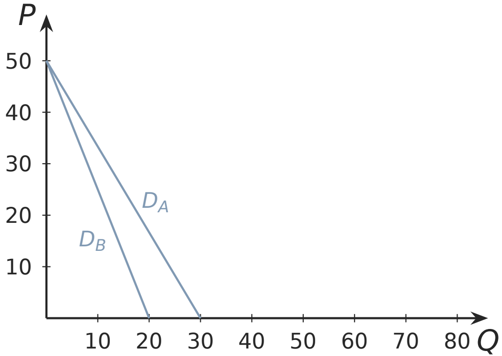
$$Q_A = 30 - 0.6P, \quad Q_B = 20 - 0.4P, \quad Q_d = Q_A + Q_B = 50 - P.$$

Market demand curve: add horizontally



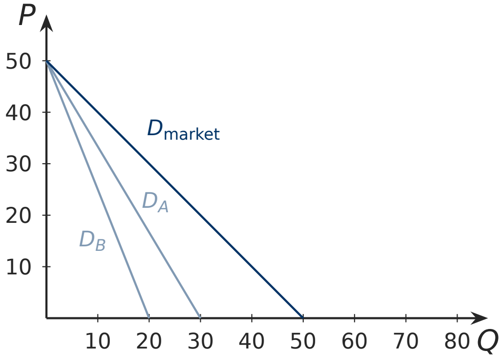
- Buyer A alone

Market demand curve: add horizontally



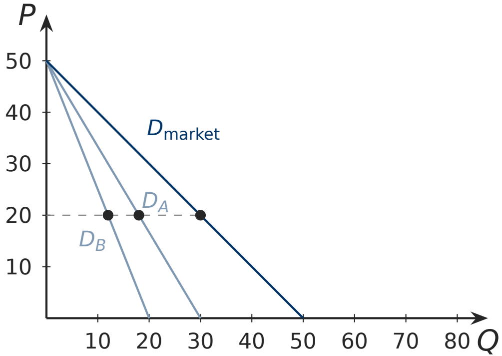
- Buyer A alone
- Buyer B alone

Market demand curve: add horizontally



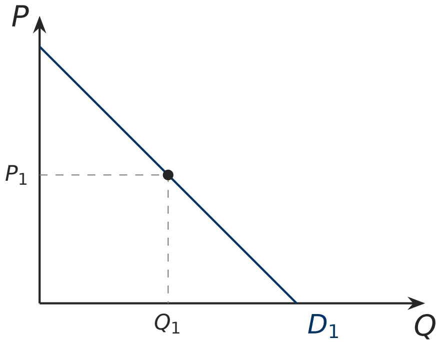
- Buyer A alone
- Buyer B alone
- Market: same price, quantities added

Market demand curve: add horizontally

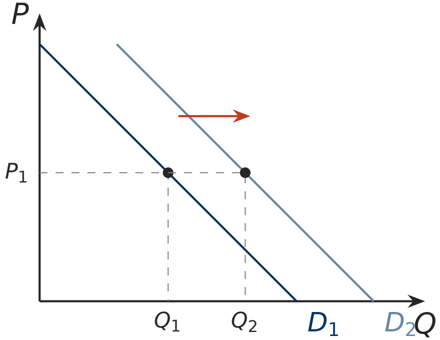


- Buyer A alone
- Buyer B alone
- Market: same price, quantities added
- At \$20: $18 + 12 = 30$

Shift of the curve vs movement along it

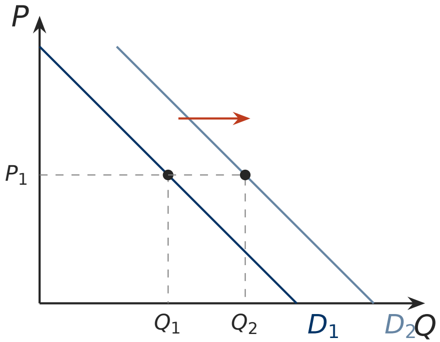


Shift of the curve vs movement along it



- Change in demand: the curve shifts. At the same price P_1, buyers want a different quantity.
- Caused by anything *other than* the good's own price.

Shift of the curve vs movement along it



- Change in demand: the curve shifts. At the same price P_1, buyers want a different quantity.
- Caused by anything *other than* the good's own price.
- Change in quantity demanded: a move along a fixed curve. Caused by the good's own price.

Rule of thumb

Variable on an axis: move along.
Variable not on an axis: shift.

What shifts the demand curve

Shifter	Change	Demand
Number of buyers	more buyers	right
Income	rises, normal good	right
	rises, inferior good	left
Related good's price	substitute gets pricier	right
	complement gets pricier	left
Tastes	shift toward the good	right
Expectations	expects higher price or income	right (today)

Reverse each change for the opposite shift. “Increase in demand” always means right.

Income: normal and inferior goods

- Normal good: income up, demand up. Restaurant meals, air travel, new cars.
- Inferior good: income up, demand down. Instant noodles, bus rides, secondhand furniture.

Normal vs inferior is about how *buyers behave*, not about the good itself. Frozen pizza may be inferior for one group and normal for another.

Question

Is a Spotify subscription a normal good or an inferior good? What would you need to observe to decide?

Prices of related goods

Substitutes

A rise in the price of one raises demand for the other. Pizza and burgers; Coke and Pepsi; Spotify and Apple Music.

Complements

A rise in the price of one lowers demand for the other. Phones and apps; printers and ink; tuition and textbooks.

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Note what moves: the *other* good's price shifts *this* good's demand curve. This good's own price never shifts its own curve.

Tastes and expectations

- Tastes. Anything that makes the good more attractive shifts demand right: advertising, a health report, fashion. Economists take tastes as given and study what happens when they change.
- Expectations. Buyers who expect a higher price next month buy more now. Buyers who expect their income to fall buy less now.
- Number of buyers. Market demand is a sum; more buyers, larger sum.

Shift or move? Spotify

Spotify charges the same price in every country it serves. Draw the demand curve for subscriptions and show what each event does to it.

Question

(a) Spotify launches in twenty more countries at the same price.

(b) A global recession lowers incomes.

(c) Apple Music cuts its price.

(d) Smartphones become cheaper.

Shift or move? Spotify, continued

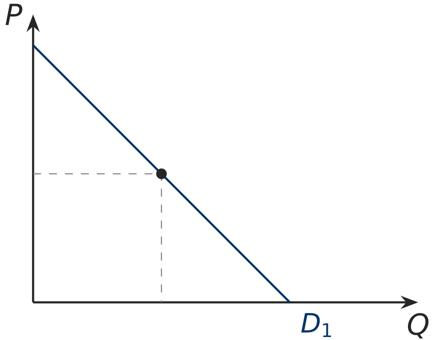
Question

(e) Spotify launches a campaign promoting local artists.

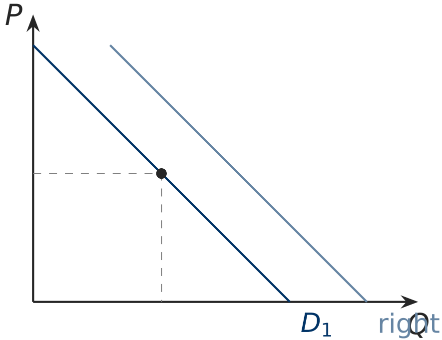
(f) Prominent economists warn that a recession is coming.

(g) Spotify ends its free tier; new users must pay the subscription price.

Spotify: answers

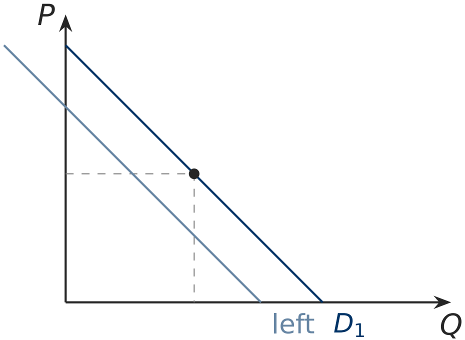


Spotify: answers



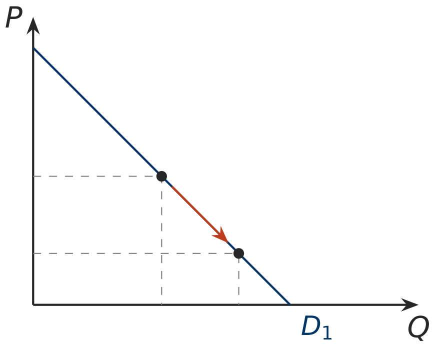
- Right: (a) more buyers; (d) cheaper complement; (e) tastes

Spotify: answers



- Right: (a) more buyers; (d) cheaper complement; (e) tastes
- Left: (b) lower income, if normal; (c) cheaper substitute; (f) lower expected income

Spotify: answers



- Right: (a) more buyers; (d) cheaper complement; (e) tastes
- Left: (b) lower income, if normal; (c) cheaper substitute; (f) lower expected income
- Movement along: (g) the price the affected users face rises from zero to the subscription price. Own price, not a shifter.

Supply

Quantity supplied

The amount sellers are willing and able to sell at a given price.

Law of supply

Other things equal, quantity supplied rises when the price rises and falls when the price falls.

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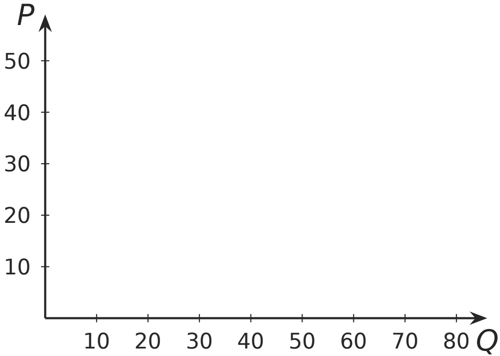
A higher price makes each unit more profitable: existing sellers produce more and marginal producers enter. The supply curve slopes up.

Supply schedule and supply curve

Price ($)	Q_s
0	0
10	15
20	30
30	45
40	60
50	75



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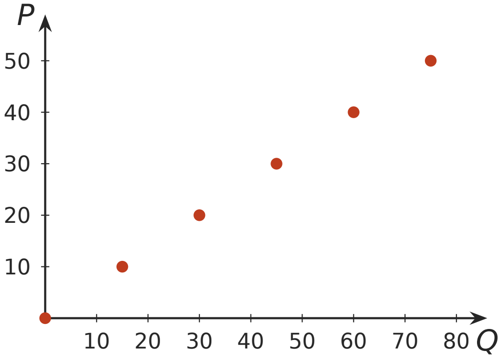

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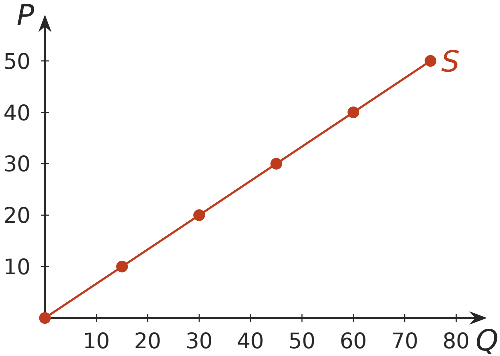
$Q_s = 1.5P$

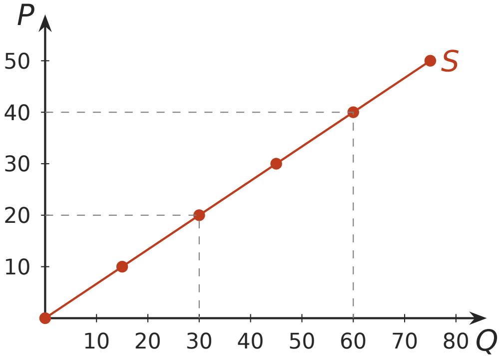


Supply schedule and supply curve

Price ($)	Q_s
0	0
10	15
20	30
30	45
40	60
50	75

$Q_s = 1.5P$

A price rise from $20 to $40 raises Q_s from 30 to 60: a movement along the curve.



Individual supply and market supply

Market supply is the sum of individual supplies: at each price, add what every seller offers.

Price ($)	Seller 1	Seller 2	Market Q_s
0	0	0	
10	10	5	
20	20	10	
30	30	15	
40	40	20	
50	50	25	

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Market supply is the sum of individual supplies: at each price, add what every seller offers.

Price ($)	Seller 1	Seller 2	Market Q_s
0	0	0	0
10	10	5	15
20	20	10	30
30	30	15	45
40	40	20	60
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0	0	0	0
10	10	5	15
20	20	10	30
30	30	15	45
40	40	20	60
50	50	25	75

$$Q_1 = P, \quad Q_2 = 0.5P, \quad Q_s = 1.5P.$$

What shifts the supply curve

Shifter	Change	Supply
Input prices	wages, materials, energy get cheaper	right
Technology	cost-saving improvement	right
Number of sellers	more firms enter	right
Expectations	sellers expect a higher future price	left (today)

- Input prices and technology work through cost: cheaper to produce, more offered at every price.
- Expectations run opposite to demand: a seller expecting a higher price holds stock back now.

Supply: shift vs movement, and two traps

- Change in supply: the curve shifts (input prices, technology, sellers, expectations).
- Change in quantity supplied: a move along the curve (the good's own price).

Trap 1

Never say supply shifts "up" or "down." An upward move of the supply curve is a *decrease*. Read horizontally: right is more, left is less.

Trap 2

The price of a substitute in consumption shifts *demand*, not supply. Cheaper orange juice changes nothing about the cost of making apple juice.

Shift or move? Supply of electric cars

Question

Draw the supply curve for electric cars. What does each event do to it?

(a) The price of lithium, used in batteries, falls.

(b) The price of electric cars falls.

(c) The price of gasoline cars falls.

Electric car supply: answers

- (a) An input got cheaper: supply shifts right.

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- (b) Own price: movement down along the supply curve. Lower Q_s, no shift.

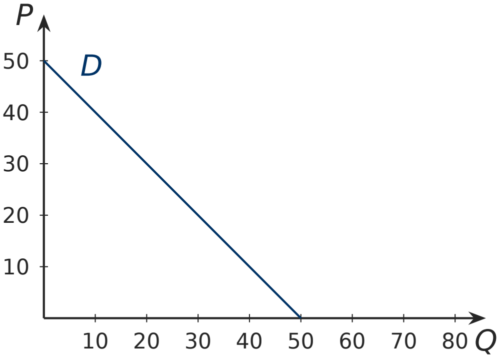
Electric car supply: answers

- (a) An input got cheaper: supply shifts right.
- (b) Own price: movement down along the supply curve. Lower Q_s, no shift.
- (c) Trick. Gasoline cars are a substitute for buyers. This shifts the demand for electric cars left. Supply does not move.

Equilibrium and changes in equilibrium

Supply and demand together

Price ($)	Q_d	Q_s
0	50	0
10	40	15
20	30	30
30	20	45
40	10	60
50	0	75



Supply and demand together

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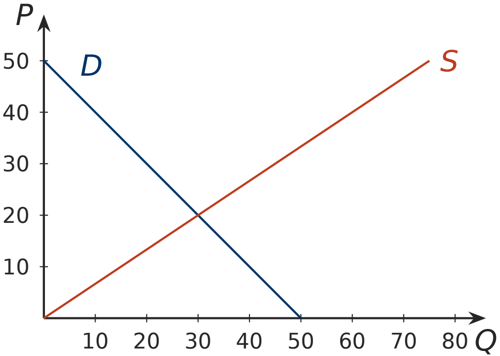
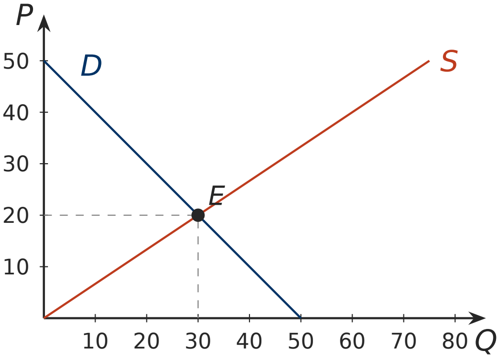


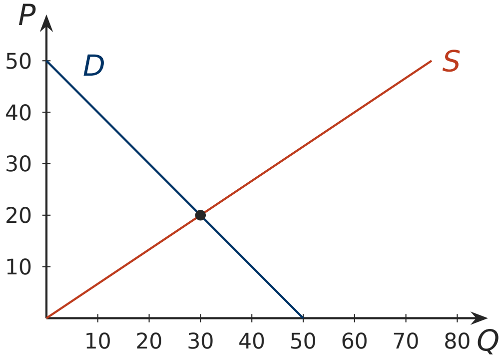
Supply and demand together

Price ($)	Q_d	Q_s
0	50	0
10	40	15
20	**30**	**30**
30	20	45
40	10	60
50	0	75

Equilibrium: $Q_d = Q_s$. Here $P^* = 20$, $Q^* = 30$.



Above equilibrium: surplus



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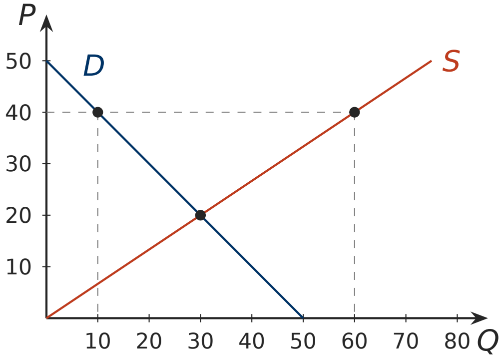
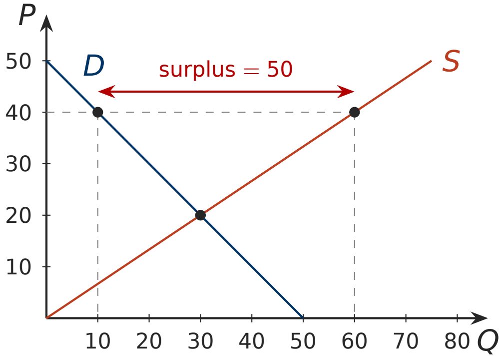


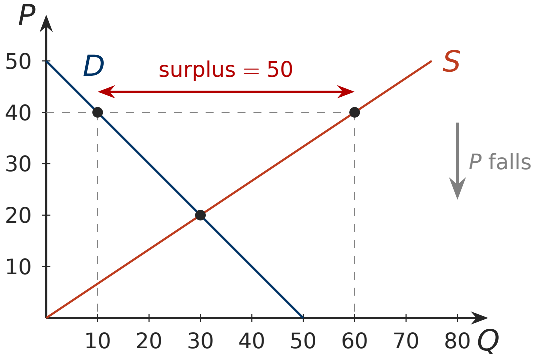
- At $P = 40$: $Q_s = 60$, $Q_d = 10$

Above equilibrium: surplus



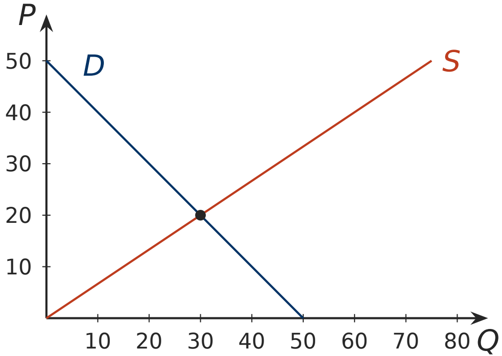
- At $P = 40$: $Q_s = 60$, $Q_d = 10$
- Surplus (excess supply) of $60 - 10 = 50$

Above equilibrium: surplus



- At $P = 40$: $Q_s = 60$, $Q_d = 10$
- Surplus (excess supply) of $60 - 10 = 50$
- Unsold stock: sellers cut the price. Q_d rises along D, Q_s falls along S, surplus shrinks to zero at \$20.

Below equilibrium: shortage



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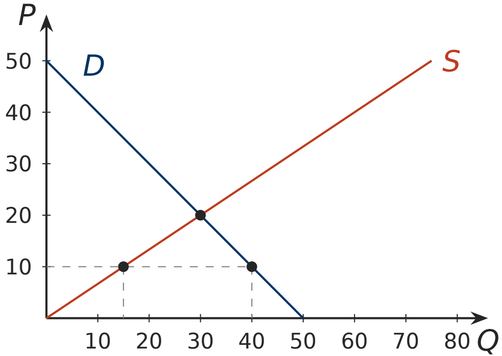
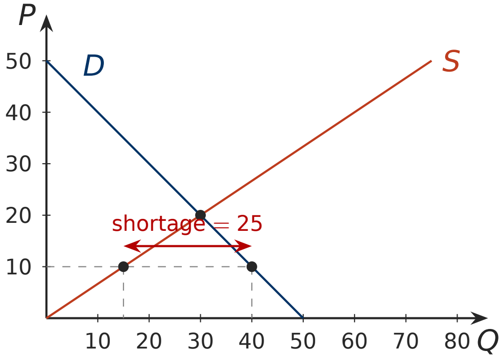


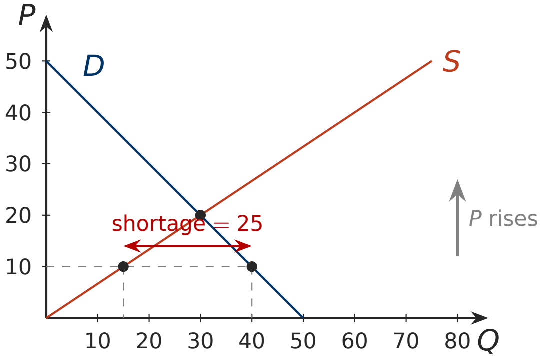
- At $P = 10$: $Q_d = 40$, $Q_s = 15$

Below equilibrium: shortage



- At $P = 10$: $Q_d = 40$, $Q_s = 15$
- Shortage (excess demand) of $40 - 15 = 25$

Below equilibrium: shortage



- At $P = 10$: $Q_d = 40$, $Q_s = 15$
- Shortage (excess demand) of $40 - 15 = 25$
- Buyers outnumber units: sellers raise the price. Q_d falls along D, Q_s rises along S, shortage closes at \$20.

The law of supply and demand

The price of any good adjusts to bring quantity supplied and quantity demanded into balance.

- Above equilibrium: surplus, price falls.
- Below equilibrium: shortage, price rises.
- At equilibrium: no pressure either way.

Speed varies by market, but in most well-functioning markets surpluses and shortages are short-lived.

Equilibrium from equations

$$Q_d = 1200 - 50P \qquad Q_s = 200 + 50P$$

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2. Solve: $\quad 1000 = 100P$, so $P^* = 10$

Equilibrium from equations

$$Q_d = 1200 - 50P \qquad Q_s = 200 + 50P$$

1. Set $Q_d = Q_s$: $\quad 1200 - 50P = 200 + 50P$
2. Solve: $\quad 1000 = 100P$, so $P^* = 10$
3. Substitute: $\quad Q^* = 1200 - 50(10) = 700$

Equilibrium from equations

$$Q_d = 1200 - 50P \qquad Q_s = 200 + 50P$$

1. Set $Q_d = Q_s$: $\quad 1200 - 50P = 200 + 50P$
2. Solve: $\quad 1000 = 100P$, so $P^* = 10$
3. Substitute: $\quad Q^* = 1200 - 50(10) = 700$
4. Check with the other equation: $\quad 200 + 50(10) = 700$ ✓

Equilibrium from equations

$$Q_d = 1200 - 50P \qquad Q_s = 200 + 50P$$

1. Set $Q_d = Q_s$: $\quad 1200 - 50P = 200 + 50P$
2. Solve: $\quad 1000 = 100P$, so $P^* = 10$
3. Substitute: $\quad Q^* = 1200 - 50(10) = 700$
4. Check with the other equation: $\quad 200 + 50(10) = 700$ ✓

Same method on the schedule: $50 - P = 1.5P$ gives $P^* = 20$, $Q^* = 30$.

Try it

Question

Demand and supply are $Q_d = 1200 - 50P$ and $Q_s = 200 + 50P$.

(a) What is the surplus or shortage at $P = 14$?

(b) Income rises and demand becomes $Q_d = 1500 - 50P$. Find the new equilibrium.

(c) At the old price of $10, what does the market look like right after the shift?

Try it

Question

Demand and supply are $Q_d = 1200 - 50P$ and $Q_s = 200 + 50P$.

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(a) $Q_d = 500$, $Q_s = 900$: surplus of 400. (b) $1300 = 100P$, $P^* = 13$, $Q^* = 850$. (c) $Q_d = 1000$, $Q_s = 700$: shortage of 300, which pushes the price up to $13.

Three steps to analyze a change in equilibrium

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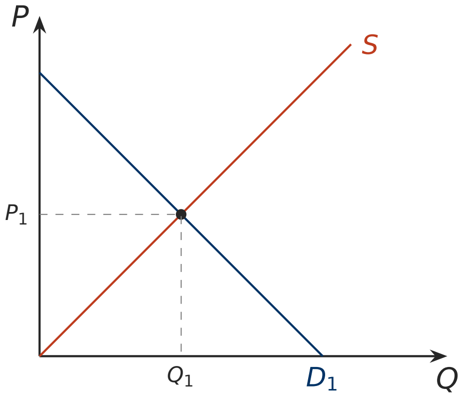
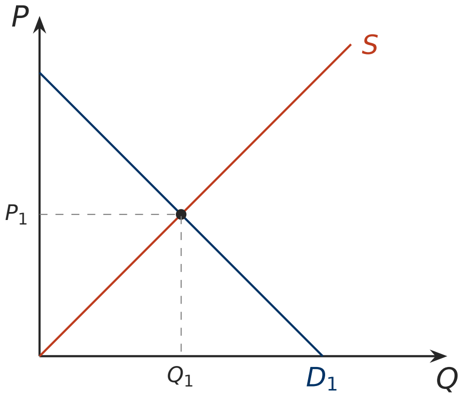
2. Does each affected curve shift right or left?
3. Draw it. Compare the new equilibrium with the old: what happened to P and Q?

Step 1 is the one students get wrong. Ask: does the event change what buyers want at a given price, or what it costs sellers to produce?

Electric cars, case A: the price of oil rises

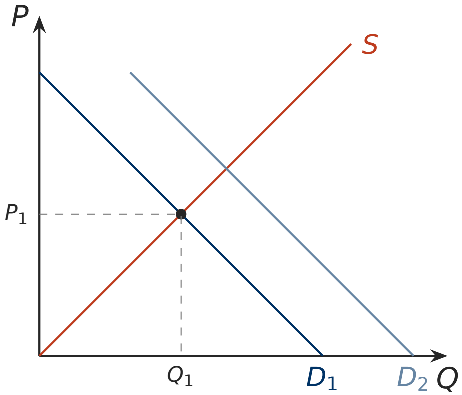


Electric cars, case A: the price of oil rises



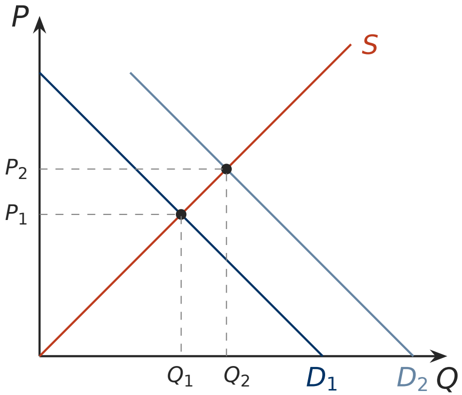
1. Gasoline cars are a substitute for buyers; they just got costlier to run. Demand shifts. (Nothing changed about building an electric car.)

Electric cars, case A: the price of oil rises



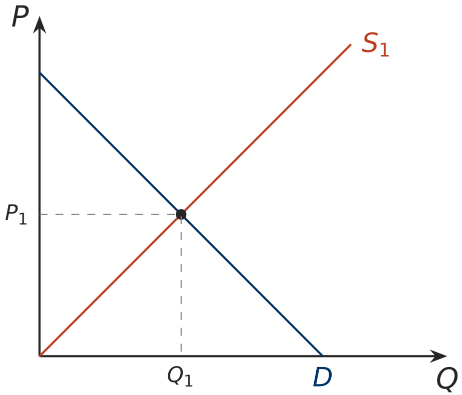
1. Gasoline cars are a substitute for buyers; they just got costlier to run. Demand shifts. (Nothing changed about building an electric car.)
2. Demand shifts right.

Electric cars, case A: the price of oil rises

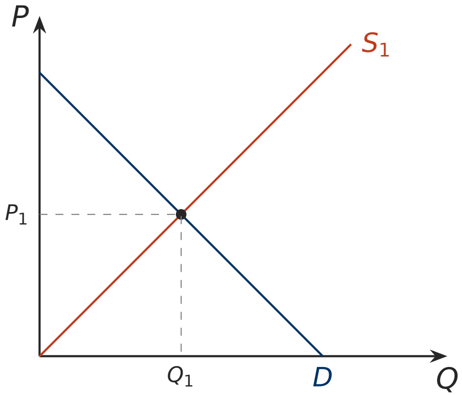


1. Gasoline cars are a substitute for buyers; they just got costlier to run. Demand shifts. (Nothing changed about building an electric car.)
2. Demand shifts right.
3. Shortage at P_1; price bid up. P up, Q up.

Electric cars, case B: engine technology improves

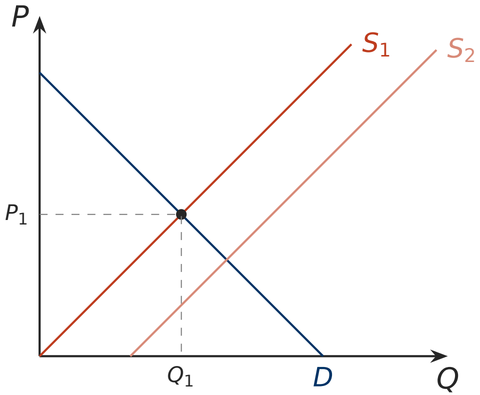


Electric cars, case B: engine technology improves



1. Lower production cost. Supply shifts.

Electric cars, case B: engine technology improves



1. Lower production cost. Supply shifts.
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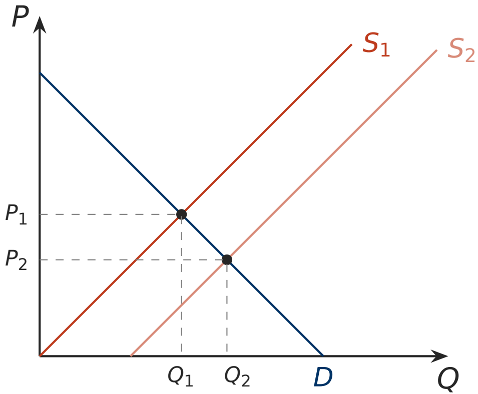
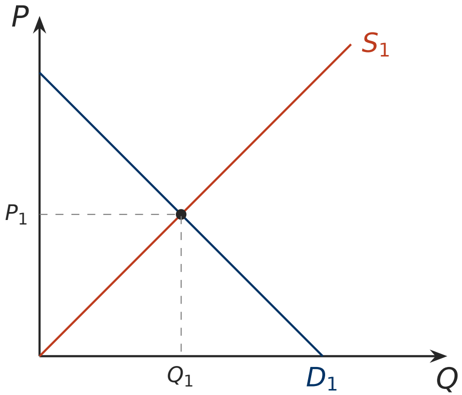
Electric cars, case B: engine technology improves



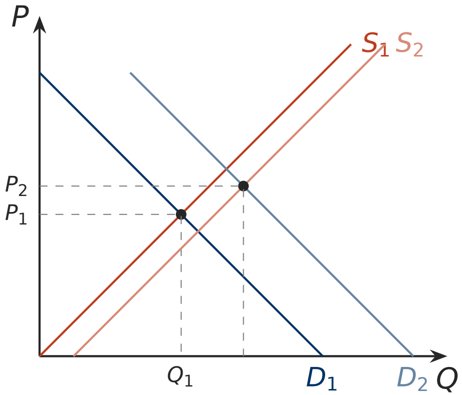
1. Lower production cost. Supply shifts.
2. Supply shifts right.
3. Surplus at P_1; price falls. P down, Q up.

Electric cars, case C: both at once



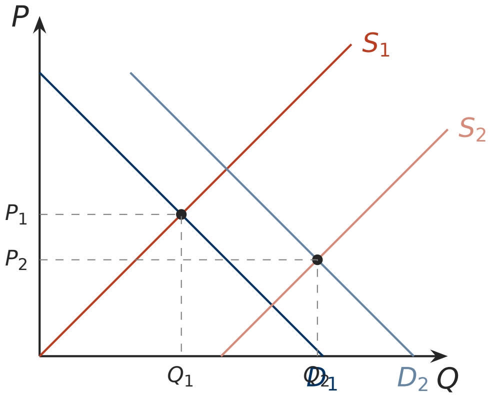
1. Both curves shift, both to the right.

Electric cars, case C: both at once



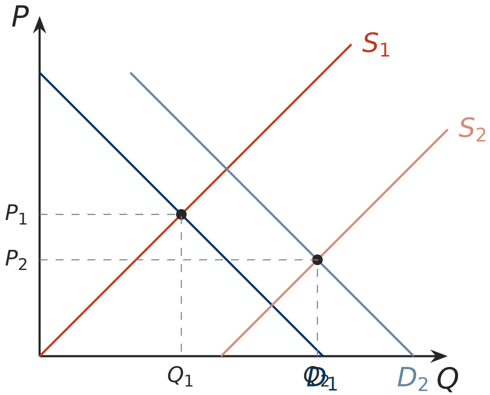
1. Both curves shift, both to the right.
2. Small supply shift: P rises.

Electric cars, case C: both at once



1. Both curves shift, both to the right.
2. Small supply shift: P rises.
3. Large supply shift: P falls.

Electric cars, case C: both at once



1. Both curves shift, both to the right.
2. Small supply shift: P rises.
3. Large supply shift: P falls.

Conclusion

Q rises for sure. P is ambiguous: it depends on which shift is larger, and the question does not say.

All nine cases

	S unchanged	S increases	S decreases
D unchanged	P same, Q same	$P\downarrow, Q\uparrow$	$P\uparrow, Q\downarrow$
D increases	$P\uparrow, Q\uparrow$	$P\,?, Q\uparrow$	$P\uparrow, Q\,?$
D decreases	$P\downarrow, Q\downarrow$	$P\downarrow, Q\,?$	$P\,?, Q\downarrow$

One curve shifts: both P and Q determined. Both shift: one determined, the other ambiguous unless the sizes are given.

Show the change consistent with the statement

Question

Draw the shift and name the shifter.

(a) Healthcare: "When there is an influx of immigrants, the price of healthcare rises."

(b) Electric cars: "When the price of electric cars is expected to rise in the near future, the current price of electric cars rises."

Show the change consistent with the statement

Question

Draw the shift and name the shifter.

(a) Healthcare: "When there is an influx of immigrants, the price of healthcare rises."

(b) Electric cars: "When the price of electric cars is expected to rise in the near future, the current price of electric cars rises."

(a) More buyers: demand right, P and Q up.
(b) Buyers expecting a higher price buy now (demand right); sellers expecting a higher price hold back (supply left). Either raises today's price; together P up, Q ambiguous.

Two markets at once

Question

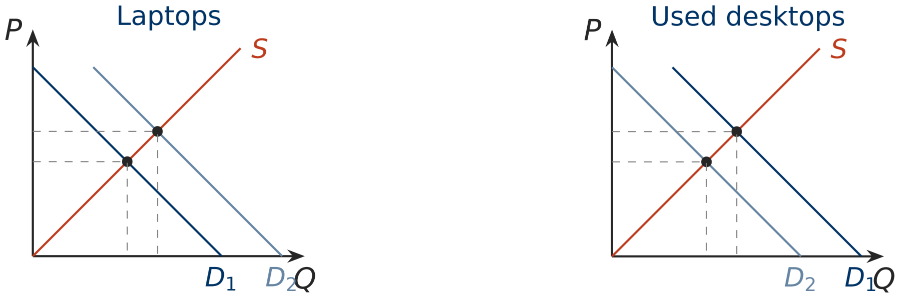
“When a technological advance improves the features of laptops, the price of laptops rises and the price of used desktops falls.”

(a) Show the change in the market for laptops.

(b) Show the change in the market for used desktops.

Hint: the statement tells you which way the laptop price moves. Use it to decide which curve shifted.

Two markets at once: answer



Laptops: a better product works through tastes; demand right, P up. (A cost-cutting advance would shift supply right and lower P, contradicting the statement.)

Used desktops: laptops are a substitute that got better; demand left, P down.

Exam-style scenarios

Question

Use the three steps. What happens to equilibrium P and Q?

(a) A hurricane destroys much of the Hudson Valley apple crop. Market for apples.

(b) A report finds dangerous mercury levels in fish; at the same time, diesel fuel for fishing boats gets cheaper. Market for fish.

(c) Honey and jam are substitutes; the price of honey rises. Market for jam. And the market for honey?

Exam-style scenarios: answers

- (a) Supply shifts left. P up, Q down. (This is what happened after Hurricane Irene in 2011.)

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Exam-style scenarios: answers

- (a) Supply shifts left. P up, Q down. (This is what happened after Hurricane Irene in 2011.)
- (b) Demand shifts left (tastes), supply shifts right (input price). Both push P down. Opposite pulls on Q: ambiguous.
- (c) Jam: demand shifts right; P and Q of jam up. Honey: its own price changed, so a movement along the honey demand curve, no shift.

How prices allocate resources

- Nobody decides how many electric cars get built or who gets them. The price does.
- Demand rises: the higher price tells producers to build more and rations the cars to the buyers who value them most.
- Costs fall: the lower price passes the gain to buyers and pulls resources into the industry.
- The equilibrium price is the one that makes plans consistent: what buyers choose to consume equals what sellers choose to produce.

This is the mechanism behind “markets are usually a good way to organize economic activity.” Later chapters ask when it fails.

Summary

- Demand curve slopes down (law of demand); shifts with buyers, income, related prices, tastes, expectations.
- Supply curve slopes up (law of supply); shifts with input prices, technology, sellers, expectations.
- Own price: movement along. Anything else: shift. Increase means right.
- Equilibrium: $Q_d = Q_s$. Above it a surplus pushes P down; below it a shortage pushes P up.
- Three steps: which curve, which direction, compare equilibria.
- One shift: P and Q both determined. Two shifts: one of them ambiguous.